experimental prospects for determining in the near future whether supersymmetry truly is a property of our world.

Of course, the reasons we have given for believing in—or at least not yet rejecting—supersymmetry are far from airtight. We have described how supersymmetry elevates our theories to their most symmetric form—but you might suggest that the universe does not care about attaining the most symmetric form that is mathematically possible. We have noted the important technical point that supersymmetry relieves us from the delicate task of tuning numerical parameters in the standard model to avoid subtle quantum problems—but you might argue that the true theory describing nature may very well walk the fine edge between self-consistency and self-destruction. We have discussed how supersymmetry modifies the intrinsic strengths of the three nongravitational forces at tiny distances in just the right way for them to merge together into a grand unified force—but you might argue, again, that nothing in the design of nature dictates that these force strengths must exactly match on microscopic scales. And finally, you might suggest that a simpler explanation for why the superpartner particles have never been found is that our universe is not supersymmetric and, therefore, the superpartners do not exist.

No one can refute any of these responses. But the case for supersymmetry is strengthened immensely when we consider its role in string theory.

Supersymmetry in String Theory

The original string theory that emerged from Veneziano’s work in the late 1960s incorporated all of the symmetries discussed at the beginning of this chapter, but it did not incorporate supersymmetry (which had not yet been discovered). This first theory based on the string concept was, more precisely, called the bosonic string theory. The name bosonic indicates that all of the vibrational patterns of the bosonic string have spins that are a whole number—there are no fermionic patterns, that is, no patterns with spins differing from a whole number by a half unit. This led to two problems.

First, if string theory was to describe all forces and all matter, it would somehow have to incorporate fermionic vibrational patterns, since the known matter particles all have spin-½. Second, and far more troubling, was the realization that there was one pattern of vibration in